

Reviewer Report (JEDC-D-26-00041):

Sorting, Price Informativeness, and Information Acquisition in Dark Markets

Paper Summary

The paper offers asset market experiments to study the interaction between endogenous information acquisition and market transparency. The authors show that the implications of market opacity on the lit markets depend on the precision of the information held by informed traders. They also show that dark trading improves market quality at moderate levels, and opaque markets are more suitable for uninformed traders.

The reported experimental results are potentially interesting; however, there are some major concerns regarding the contribution and exposition of the current draft.

Comments

Novelty and Contribution. One of the key findings of the paper is that there is an increase in information acquisition thanks to dark trading when information precision is high. How is this finding different than the observation made by Brogaard and Pan (2022) around earnings announcements? The authors could also refer to the papers on venue choice in fragmented markets (e.g., Menkveld, Yueshen and Zhu, 2017), which emphasize the trade-off between the cost savings in dark pools and immediacy due to execution risk in opaque markets. The authors mainly focus on the execution risk, but the cost aspect is less clear in the paper. The authors report only gross profits, but net profits of informed traders would be informative about the value of acquiring information.

In the experimental design, a large fraction of traders are informed. There are at least four uninformed (noise) traders, yet nearly half of the trading volume comes from the uninformed traders. What is their incentive to trade? **Is there any difference between exogenously and endogenously uninformed traders in terms of trading strategies?**

More importantly, since there are no liquidity traders in the market with varying trading targets, e.g., large institutional trades, as in other experimental studies on market microstructure (e.g., Bloomfield et al., 2015), the design does not allow to study the interaction of trade size with venue choice or the competition in liquidity provision (Boulatov and George, 2013; Foley and Putnins, 2016), important dimensions of routing decisions. Do

dark traders still benefit small uninformed (noise) traders in the presence of large uninformed institutional traders?

Exposition. The authors offer too many results, yet a focused discussion on the key results and the underlying mechanisms, citing the relevant literature, would improve the paper. For example, what are the new dimensions of the first two results that have not been studied theoretically, empirically, and in experimental markets (Hendershottt, Wee and Wen, 2022). For example, are the first two findings due to only adverse selection, trading strategies around limit versus market order submissions or trade-throughs, why do traders with a strong signal prefer lit market when precision is high (results 3) despite higher execution costs and loss of informational rent, why does the addition of a dark pool does not affect price quality (result 4) while increasing information acquisition, what is the trade-off driving the non-linear relationship (result 5), the probability of execution vs. the size of the information rent, why do uninformed earn more in dark pools (result 8), due to cost savings in dark pools or escaping adverse selection on lit exchanges, or both, does the existence of a dark pool affect the decision to acquire information in the first place? Clearer answers to these questions would improve the exposition and highlight the novelty of the findings.

The non-linearity result on dark participation and price efficiency could be better motivated. Although it is already documented empirically (Comerton-Forde and Putnins, 2015), the authors frame their findings within the context of MiFIR changes around double volume caps (DVC) implemented in Europe. The findings of the authors could potentially contribute to the regulatory discussion, but they should be interpreted with some caution, as important dimensions of the problem are lacking in the current experimental design, as described above.

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